



2026–2027



جب کوئی قوم فن اور علم
سے عاری ہو جاتی ہے
تو وہ غربت کو دعوت
دیتی ہے اور جب غربت
آتی ہے تو وہ ہزاروں
جرائم کو جنم دیتی ہے۔

“When a nation becomes devoid
of art and learning, it invites
poverty and when poverty comes
it brings in its wake thousands of
crimes.”

– Sir Syed Ahmad Khan

Lab-I

Course Code- CAMS1P01



MCA I Semester Lab Manual

DEPARTMENT OF COMPUTER SCIENCE
ALIGARH MUSLIM UNIVERSITY, ALIGARH

2026–2027

Credits

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Revised Edition: July 2026

Department of Computer Science, A.M.U., Aligarh (U.P.) India

Department of Computer Science

Aligarh Muslim University, Aligarh

MCA I Semester

Lab Manual

Laboratory Course-I (CAMS-1P01)

Fundamentals of IT, C++ Programming and Database Practice

Credit: 4

Periods per Week: 6

Continuous Assessment: 40 Marks
60 Marks

Exams:

CONTENTS

Week	Laboratory Focus	Page
1	Fundamentals of IT (MS Word and MS Excel)	8
2	Fundamentals of IT (MS PowerPoint)	13
3	C++ Environment, IDE Installation and Tool Setup	14
4	C++ Operators, Decision-Making and Control Statements	25
5	One- and Two-Dimensional Arrays in C++	27
6	Pointers and Pointer Operations in C++	29
7	Pointers and the this Pointer	31
8	Strings and String Operations in C++	33
9	Functions, Recursion and Relational Database Queries	34
10	Structures in C++ and Database Queries	36
11	Object-Oriented Programming, Classes and Objects	38
12	Constructors, Destructors and Database Design	40
13	Operator Overloading, Type Conversion and Inheritance	42
14	Binding, Virtual Functions, Streams and Exception Handling	44

COURSE TITLE:	LABORATORY COURSE-I	COURSE CODE:	CAMS-1P01
CREDIT:	4	PERIODS PER WEEK:	6
CONTINUOUS ASSESSMENT:	40 Marks	EXAMS:	60 Marks

COURSE DESCRIPTION

Knowledge of a programming language is a prerequisite to the study of most computer science courses. This knowledge area consists of those skills and concepts that are essential to problem-solving and programming practice, independent of the underlying paradigm. All programming languages will share one similarity, i.e., all are based on logic. A programming language is a programmer's principal interface with the computer. More than just knowing how to program in a single language, programmers need to understand the different styles of programming promoted by different languages. Later in your professional life, you will be working with many different programming languages and styles at once and will encounter many different languages over the course of your career. Understanding the variety of programming languages and the design trade-offs between the different programming paradigms makes it much easier to master new languages quickly.

This course examines practical programming techniques and issues, emphasizing object modelling and simulation. The objectives of the course are to explore issues involved in developing large-scale object-oriented systems and to teach fundamental techniques that can simplify software development. The course provides in-depth information on object-oriented programming, issues, techniques, and methodologies.

CONTENT

This course introduces fundamental 'computer literacy' concepts. The objective is to explore the knowledge of the fundamentals of information technology and information systems.

This course is designed to provide the students with the opportunity of learning both concepts of C++ and then learning object-oriented concepts using C++. Our goal will be to improve your reasoning and thinking skills, which should prove helpful not only in future programming but throughout your academic and professional career.

OBJECTIVES

This course is designed:

- ▶ To help the students in learning C++.
- ▶ To help the students write a sample program in C++.
- ▶ To help the students understand the use of different variables.

- ▶ To help the students understand the use of decision-making statements.
- ▶ To be able to debug and test C++ programs.
- ▶ Understand how to read C++ library documentation & reuse library code.
- ▶ To make the students understand the features of object-oriented principles and familiarize them with virtual functions, templates, and exception handling.

OUTCOMES

After completing this course, the students would be able to:

- ▶ Write, debug, and run a sample program in C++.
- ▶ Differentiate the variables and constants.
- ▶ Understand how decision-making statements are written.
- ▶ Create a simple real-life application based on the decision-making statements.
- ▶ Understand the concepts and implementation of constructors and destructors.
- ▶ Develop software applications using an object-oriented programming language in C++.
- ▶ Understand and use the basic programming constructs of C++.
- ▶ Learn C++ data types, memory allocation/reallocations, functions, and pointers.
- ▶ Apply object-oriented programming concepts to software problems in C++.

HOW TO DO WELL IN THIS COURSE

- ▶ The students are advised to attend all their theory classes and respective labs regularly, as both are integrated into each other. If any student misses the theory lecture, he/she may not be able to do well in the lab related to that topic.
- ▶ The students are advised to submit the assignments given in theory and lab classes in a timely manner to their respective Teachers/Instructors online.
- ▶ The students should demonstrate a disciplined and well-behaved demeanour in the department.
- ▶ Each student shall be assigned a system in their introductory lab. They are advised to do their work on that system only for the whole semester. Students should store all their lab activities regularly.
- ▶ All students are advised to understand course objectives and outcomes and achieve both during their lab work.
- ▶ The students are advised to follow the books/eBooks/online tutorials/other online study material links given in the lecture/lab manual/syllabus references. These study materials are very helpful in terms of skills, knowledge, and placement.
- ▶ This Lab course is very important in terms of placement. Therefore, students are advised to implement all the problems by her/him given in the individual week.
- ▶ All students are advised to solve old placement papers for campus selection.

Placement Preparation Links

- ▶ <https://www.indiabix.com/placement-papers/companies/>
- ▶ <https://www.offcampusjobs4u.com/download-tcs-placement-test-question-papers-with-solutions/>
- ▶ <https://www.indiabix.com/placement-papers/tcs/>
- ▶ <https://www.firstnaukri.com/career-guidance/infosys-placement-papers-with-solutions-2019-firstnaukri-prep>
- ▶ <https://prepinsta.com/ibm/>
- ▶ <https://www.faceprep.in/infosys/infosys-aptitude-questions/>
- ▶ <https://alpingi.com/infosys-placement-papers-solution-pdf-download/>
- ▶ <http://placement.freshersworld.com/>

Tutorial and Installation Links

- ▶ <https://www.javatpoint.com/cpp-tutorial>
- ▶ <https://www.tutorialspoint.com/cplusplus/index.htm>
- ▶ <http://www.cplusplus.com/doc/tutorial/>
- ▶ <http://www.bloodshed.net/dev/devcpp.html>
- ▶ <http://www.codeblocks.org/downloads>

Online C++ Editors

- ▶ J.D.O.O.D.L.E. Online C++ Compiler: <https://www.jdoodle.com/online-compiler-c++/>
- ▶ Online GDB Online C++ Compiler: https://www.onlinegdb.com/online_c++_compiler
- ▶ C++ Shell Online C++ Compiler: <http://cpp.sh/>
- ▶ Programiz C++ Online Compiler: <https://www.programiz.com/cpp-programming/online-compiler/>

RULES AND REGULATIONS

Students are required to strictly adhere to the following rules.

- ▶ The students must complete the weekly activities/assignments well in time (i.e., within the same week).
- ▶ The students must maintain the Lab File of their completed activities/assignments in the prescribed format (Appendix-1).
- ▶ The students must get the completed weekly activities/assignments checked and signed by the concerned teachers in the lab in the immediately succeeding week. Failing which the activities/assignments for that week will be treated as incomplete.
- ▶ At least TEN (10) such timely completed and duly signed weekly activities/assignments are compulsory, failing which students will not be allowed to appear in the final Lab Examination.
- ▶ The students need to submit the following three deliverables for each exercise, duly signed by the Teacher:
 - Coding
 - Input /Output
- ▶ The students need to ensure that each question is assessed and signed by the Teacher in the week/time.
- ▶ Late submission would not be accepted after the due date.
- ▶ Cooperate, collaborate, and explore for the best individual learning outcomes, but copying is strictly prohibited.

APPENDIX-I**Template for the Index of Lab File**

Week	No.	Problems with Description	Page No.	Teacher's Signature with Date
1	1#			
	2#			
	3#			
2	1#			
	2#			
	3#			
3	1#			
	2#			
	3#			

Note: The students should use Header and Footer mentioning their roll no. & name in the footer and page no. in the header.

WEEK #1**Fundamentals of IT: MS Word and MS Excel****OBJECTIVES**

- ▶ To help the students in learning the practical use of MS-Word, MS-Excel.

OUTCOMES

After completing this, the students would be able to:

- ▶ Use MS-Word, MS-Excel in real-life applications.

PROBLEMS

Problem 1. Open a new document and type the following letter.

July 15, 2020,
Chennai

From

VENKATESH. P
Sri Ranga Apartments,
No: 120, II Avenue,
T. Nagar. Chennai-17

To

<<Name>>

<<Address>>

Respected <<Name>>

With the current slowdown in hiring within the high-tech field, you must be flooded with resumes from out-placed software engineers such as me. Please take a moment to consider my qualifications. I believe in particular is highly marketable in this tight market:

I worked on the team that pioneered the technology that put the Palm Pilot on the map. In today's increasingly mobile society, this technology has places to go, and I have ideas that could take us to the next step in office independence.

Please call me with prospective job opportunities. I am interested in a project management position in the Rs. 9K range.

Thank you!

Venkatesh.

Enclosure: Resume (the format given below).

1. Save the document as “Letter.doc.”
2. Send the document to 3 recipients using Mail merge. (Use 3 different addresses)
3. Define a Macro “Decorate” which makes the text bold, Red in color and italic, font size Assign a shortcut key Alt + Z to this macro.
4. Close the document.
5. The Sample Addresses are:
 - a. Mr. Amit Tandon, 13, New Estate, Ring Road, Chandigarh
 - b. Mr. Rohit Saluja, 15, Karol Bagh, New Delhi
 - c. Ms. Jyoti Parmar, Sector 16, New Building, Gurugram

Format of Resume

Name: _____

Father's Name: _____

Date of Birth: _____

Age: _____

Address: _____

Educational Qualification:

Sr.	Qualification	Board/University	Percentage
1			
2			
3			

Work Experience: _____

Technical Skills: _____

Personal Skills: _____

Hobbies: _____

Dated: _____

Signature _____

Problem 2. Create a table in Word as shown below:

Roll No	Name	Marks in Physics	Marks in Chemistry	Total Marks
1	Sakshi	80	70	
2	Rohit	70	80	
3	Amit	60	50	
4	Rakesh	40	60	
5	Komal	30	70	
6	Garima	80	80	

- ▶ In the total marks' column, entries should be calculated using formulas and it is the sum of marks in physics and marks in chemistry.
- ▶ Insert a new row at the end of the table and also find grand total using formula.
- ▶ Sort the table based on total marks.
- ▶ The date and heading should be centre aligned.
- ▶ Heading should be in bold and underlined.

Problem 3. Using a spreadsheet package you have studied, construct T Morongo's pay slip for December 2016 following the instructions below. Insert a custom footer with your name, subject, course, exam/Test & question number. Save it as Salary advice.

SALARY ADVICE FOR MARCH 2016

EMPLOYEE	T MARONGO	STAFF NO	004
DATE	31 MARCH 2016	NEXT PAY DATE	30 A.P.R.I.L. 2016
BASIC SALARY p.a.	31200.00		
INCOME	AMOUNT	DEDUCTIONS	AMOUNT
Basic Salary		Pension @8%	
Housing Subsidy		P.A.Y.E.	
Vehicle Allowance		U.I.F.	
Medical Aid		Bond Repayment	
Gross Income		Total Deductions	
		Net Salary	

Instructions

1. Housing Subsidy 6000.00 per year.
2. Car Allowance 100.00 per month.
3. Pension 8% on Basic Salary.
4. PAYE 636.83.
5. Medical Aid 70.00.
6. U.I.F. 1% on Basic Salary + Housing Subsidy.
7. Bond Repayment 630.00.
8. Calculate Net Salary.
9. Format all figures to two decimal places and insert ₹ currency symbol.
10. Insert a custom footer with your name, subject, and question number. Save it as salary advice2.

Problem 4. Use a new workbook & construct a worksheet with the data given & save it as LYONS.

LYONS INC – Orange JUICE Sales

PRODUCT	COST PRICE PER LITRE	MARK UP PER ITEM 35%	SELLING PRICE	LITRES SOLD	TOTAL INCOME	PROFIT
Cascade	3.75			234		
Quench	3.65			345		
Xtra	4.25			456		
Sun Splash	1.50			123		
House Brand	1.50			245		
TOTAL						
HIGHEST						
LOWEST						

Instructions

- ▶ Markup = Cost price/Litre x 35%
 - ▶ Selling price = Cost price/Litre + Mark up
 - ▶ Total income = Litres sold x Selling Price
 - ▶ Profit = Total income - (Cost price/Litre x Litres sold)
- a) The MARKUP % (35%) must be inserted in a separate cell under the heading. USE IT as an absolute cell reference in the formula to calculate the mark up per item.
 - b) Calculate the mark up for each item.
 - c) Calculate the selling price for each item.
 - d) Calculate the Total Income for each item.
 - e) Calculate the profit for each item.
 - f) Format the column LITRES SOLD to display the number of litres as integers. The rest of the worksheet must be formatted to display two decimals.
 - g) Use statistical functions to calculate the AVERAGE, HIGHEST (MAX.) and LOWEST (MIN) for Selling Price column up to Profit Column.
 - h) Show all formulas you have used in a new sheet. Adjust the column width so that the formulae are displayed in full and the sheets fits into one side of A4 landscape format and save it as formulas.
 - i) Under the worksheet Create a pie chart titled PRODUCT COST PER UNIT for Product & Cost price per Litre columns. Data labels indicating percentages should be displayed.
 - j) Put borders neatly on the on the work sheet & save it as LYONS2.

WEEK #2**Fundamentals of IT: MS PowerPoint****OBJECTIVES**

- ▶ To help the students in learning about MS-Power Point.

OUTCOMES

After completing this, the students would be able to:

- ▶ Understand the usage of MS-Power Point in real life.

PROBLEMS

1. Design Seasonal Greeting cards using MS-PowerPoint.
2. Design an AMU Magazine cover in MS PowerPoint. Use the following:
 - i) Select a theme for the page.
 - ii) Insert either a picture or clipart.
 - iii) Use WordArt.
3. Design a poster inviting all students of your department to the IT Fest (using MS-Power Point).
4. Create a 5-slide presentation on any topic. Use Images, Graphs, Chart, Tables, Animation, Time, Bullets, Transition, Sound, Hyperlink, Background template, Header and Footer (using MS-Power Point).
5. Create a 5-slide presentation on any topic. Use Images, Graphs, Chart, Tables, Animation, Time, Bullets, Transition, Sound, Hyperlink, Background template, Header and Footer (using MS-Power Point).

WEEK #3**C++ Environment, IDE Installation and Tool Setup****OBJECTIVES**

- ▶ To help the students in learning the concepts of C++.
- ▶ To help the students in learning the different tools for C++ programming on different software platform.
- ▶ To help the students install the C++ toolkit on their computers.

OUTCOMES

After completing this, the students would be able to

- ▶ Understand the concepts of C++.
- ▶ Setup the C++ environment on a different platform.

Dev C++ IDE: Installation, Features, and C++ Development

Dev-C++ is a fully featured graphical IDE (Integrated Development Environment) that uses the MinGW compiler system to create Windows as well as Console based C/C++ applications. It can also be used with any other GCC-based compiler like Cygwin.

Dev-C++ is free software and is distributed under the GNU General Public License. Thus, we can distribute or modify the IDE freely. It was originally developed by "Bloodshed Software". It has been formed by Orwell after it was abandoned by Bloodshed in 2006.

Section 1.01 Features of Dev-C++ IDE

Enlisted below are some of the features of this IDE that help us developing efficient and user-friendly C/C++ applications.

- ▶ Dev-C++ supports GCC-based compilers including Cygwin, MinGW, etc. We can either install a Dev-C++ IDE along with the compiler integrated or just an IDE if we already have a compiler on our system.

- ▶ We can use integrated debugging (using GDB) with this IDE. The debugger allows us to perform all the general debugging operations on source code.
- ▶ It has a localization feature that provides support for multiple languages. We can select the language the first time when we open the IDE after installing it. We can also change the language anytime using settings.
- ▶ Like the other IDEs, this IDE also provides the "Auto-Completion" feature for the code we write.
- ▶ It comes with customizable syntax highlighting editor that can make the source code more readable.
- ▶ Allows editing and compiling the Resource files.
- ▶ Has a Tool Manager that contains various tools that can be used in the project.
- ▶ This IDE also has inbuilt Find and replace facilities.
- ▶ Using Dev-C++ IDE, we can create various types of applications be it Windows, Console, Static libraries or DLLs.
- ▶ We can also create our own project templates to create our own project types.
- ▶ Make files that are used for managing the build process for the application can also be created using dev-C++ IDE.
- ▶ It provides support for Class Browser as well as Debug variable Browser.
- ▶ It has a Project Manager that helps us to manage various projects.
- ▶ Also provides print support through its interface.
- ▶ We can easily install the add-on libraries using the package manager provided by the IDE.
- ▶ This C++ IDE also provides CVS support for source code management.

Section 1.02 Installing and Configuring C++ IDE

We can get the appropriate installable for dev-C++ IDE from <http://www.bloodshed.net/dev/devcpp.html>. Let's see the entire installation process now. We have used the installable that comes along with the C++ compiler. In this tutorial, we use the dev-C++ version 5.11 with the TDM-GCC 4.9.2 compiler. The stepwise installation for dev-C++ is given below.

1. The first step while we start the installer is to select the language of our choice as shown in the below screenshot.
2. Once you select the appropriate language, you have to agree to the license agreement that pop-ups next.
3. Next, we are asked to select the components that we need to install as a part of the dev-C++ installation. As shown in the above screenshot, we are provided with a list of components available for installation and a checkbox against each component. We can

check/uncheck each box to indicate which components to install. Click next once the components are selected.

4. Now the installer prompts the user for the destination folder where the dev-C++ files/libraries etc. are to be copied. Once we provide the destination folder path, click on Install.
5. The following screenshot shows the progress of the installation. Once the installation is over, a "finish" dialog that signals the end of the installation appears. We click finish and then we can launch the dev-C++ IDE. Now let's see the working of this C++ IDE in detail.

(a) Development Using Dev-C++ IDE

(b) Configuring Dev C++

(c) Change Linker Setting for Debugging

After starting the IDE, the first thing we need to ensure is the setting for debugging information to be generated. Follow the steps below to set the debugging information.

- ▶ To change this setting, click on Tools → Compiler Options.
- ▶ Then click on the "Settings" tab on the dialog that pops up.
- ▶ Under "Settings", we have a "linker" tab.
- ▶ In the "linker" tab there are various options shown. Set "Yes" for the option "Generate Debugging Information (-g3)". This is shown in the following screenshot. Click OK, once done.

(d) Create a New Project

To create a new project in dev-C++ we need to follow the below steps

- ▶ Click File → New → Project.
- ▶ A new dialog opens up as shown below.
- ▶ Here, we can specify the project name. Make sure to select the "Empty Project" and also to check the "C++ Project" button.
- ▶ Once the entire information is provided, we can click ok and the IDE will ask for the path where the project is to be saved. When this is done, a workspace will open with the project explorer on the left-hand side that shows the project we just created.

- Now we can add or import the code files into this project.

(e) Add Source File(s)

Adding a file to a project can be done in two ways. 1. Add a new file by clicking Project → New File or Right-click on Project Name in the project explorer and click New File. 2. Another way is to add the existing files to the project. This can be done by clicking Project → Add to Project or right-click on Project Name in the project explorer and select "Add to Project..." This will give a dialog to select files and import them to the project. 3. Once the files are added to the project, the workspace looks as shown below.

(f) Compile/Build & Execute Project

When we have all the code ready for the project, we will now compile and build the project. Follow the below steps to build and execute the dev C++ project:

- To compile the project, click Execute → Compile (or click F9).
- We can see the compilation status in the "Compile Log" tab in the workspace.
- If there are any errors whether syntax or linker errors, then they will appear in the compiler tab.
- Once the project is compiled successfully, we need to run it.
- Click on Execute → Run. (or click F10)
- The console window that gives us the output will be shown in the below screenshot.
- If there are command line parameters to be passed to the program, we click on Execute → Parameters. This will open a dialog using which we can pass parameters.

(g) Debugging In C++ IDE

Sometimes we may not get the desired output from our program although the program is syntactically correct. In such a situation, we can debug the program. The

dev-C++ IDE provides the inbuilt debugger. Follow the below steps to debug the program using Dev-C++ IDE:

- Click Execute → Debug. (or click F5).
- Once the debug is clicked, we get the debug menu in the IDE, as shown below.
- Before debugging we can toggle breakpoints using F4 at a particular line of code.
- Using the debug menu, we can use options like add watches, run to cursor, into function,

etc. to efficiently debug our program.

Article II. Setting up Code::Blocks on Windows

This tutorial gives you easy-to-follow instructions, with screenshots, for setting up a compiler (the MINGW compiler), a tool that will let you turn the code that you write into programs, and Code::Blocks, a free development environment for C and C++. This tutorial explains how to install Code::Blocks on Windows 2000, XP, Vista or Windows 7.

Section 2.01 Step 1: Download Code::Blocks

- ▶ Go to this website: <http://www.codeblocks.org/downloads>
- ▶ Follow the link to "Download the binary release" (direct link)
- ▶ Go to the Windows 2000 / XP / Vista / 7 section
- ▶ Look for the file that includes mingw in the name. (The name as of this writing was codeblocks-10.05mingw-setup.exe; the 10.05 may be different).
- ▶ Save the file to your desktop. It is roughly 74 megabytes.

Section 2.02 Step 2: Install Code::Blocks

- ▶ Double click the installer.
- ▶ Hit next several times. Other setup tutorials will assume you have installed in C:\Program Files\CodeBlocks (the default install location), but you may install elsewhere if you like
- ▶ Do a Full Installation
- ▶ Launch Code::Blocks

Section 2.03 Step 3: Running in Code::Blocks

You will be prompted with a Compilers auto-detection window: When you get the compiler auto-detection window, just hit OK. Code::Blocks may ask if you want to associate it as the default viewer for C/C++ files—I'd suggest you do. Click on the File menu, and under "New", select "Project..." The following window will come up: Click on "Console Application" and hit the "Go" button. Click next until you get to the Language Selection Dialog: You'll be asked to choose whether you want to use C or C++. If you're not sure, use C++. Otherwise, choose based on the language you are learning. After clicking "Next", Code::Blocks will then prompt you with where you'd like to save the console application: I'd recommend you put it in its own folder, as it may create several files (this is especially true if you create other

types of projects). You will need to give your project a name, anything will be fine. Clicking "Next" again will prompt you to set up your compiler: You don't need to do anything here. Just accept the defaults by hitting "Finish". You can now open the main.cpp file on the left: (You may need to expand the contents of the "Sources" folder if you don't see main.cpp.) At this point, you will have your main.cpp file, which you can modify if you like. For now, it just says "Hello World!", so we can run it as is. Hit F9, which will first compile it and then run it. You now have a running program! You can simply edit main.cpp and then hit F9 to compile it and run it again.

```
sudo apt-get update
```

```
sudo apt-get install GCC
```

```
sudo apt-get install build-essential
```

```
g++ --version
```

Linux Installation: Install GCC the C++ compiler on Ubuntu 18.04

Bionic Beaver Linux We will install the GNU GCC compiler on Linux. To install and work with the GCC compiler on your Linux machine, proceed according to below steps:

- ▶ You have to first run the below two commands from your Linux terminal window: This command will install the GCC compiler on your system. You may also run the below command: This command will install all the libraries which are required to compile and run a C++ program.
- ▶ After completing the above step, you should check whether the GCC compiler is installed in your system correctly or not. To do this you have to run the below-given command from Linux terminal:
- ▶ If you have completed the above two steps without any errors, then your Linux environment is set up and ready to be used to compile C++ programs. In further steps, we will learn how to compile and run a C++ program on Linux using GCC compiler.
- ▶ Write your program in a text file and save it with any file name and .CPP extension. We have written a program to display "Hello World" and saved it in a file with the filename "helloworld.cpp" on desktop.

```
g++ filename.cpp -o any-name
```

```
g++ helloworld.cpp -o hello
```

```
./hello
```

- ▶ Now you have to open the Linux terminal and move to the directory where you have saved your file. Then you have to run the below command to compile your file: `file-name.cpp` is the name of your source code file. In our case, the name is "helloworld.cpp" and any-name can be any name of your choice. This name will be assigned to the executable file which is created by the compiler after compilation. In our case, we choose any-name to be "hello". We will run the above command as:
- ▶ After executing the above command, you will see a new file is created automatically in the same directory where you have saved the source file and the name of this file is the name you chose as any-name. Now to run your program you have to run the below command: This command will run your program in the terminal window.

IDE INSTALLATION SCREENSHOTS

The following screenshots are retained from the original manual as visual references for the Dev-C++ and Code::Blocks installation and configuration procedures.

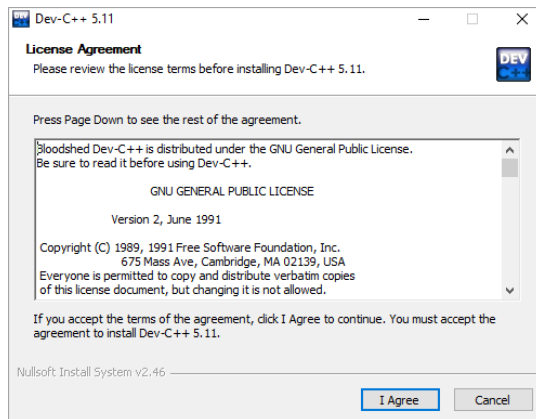


Figure 1: Dev-C++ licence agreement.

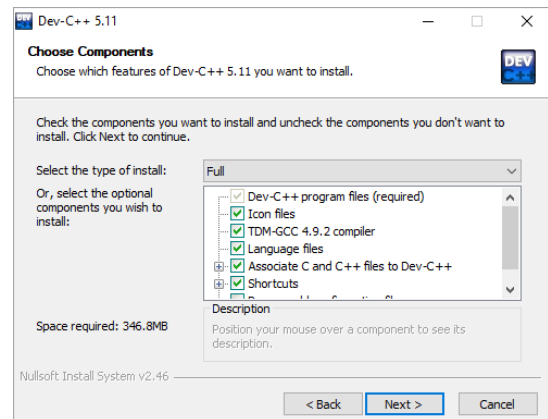


Figure 2: Dev-C++ component selection.

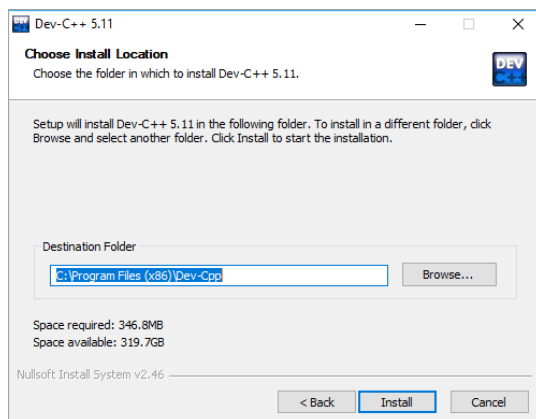


Figure 3: Destination folder selection.

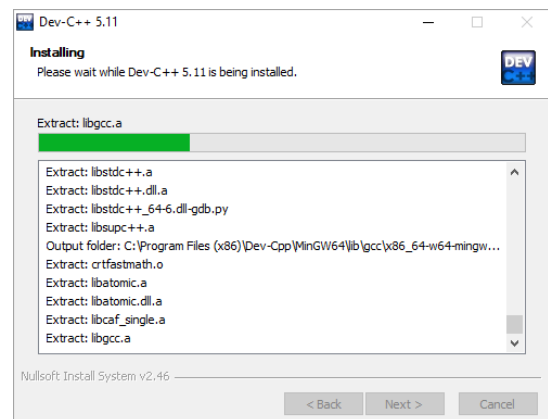


Figure 4: Installation progress.

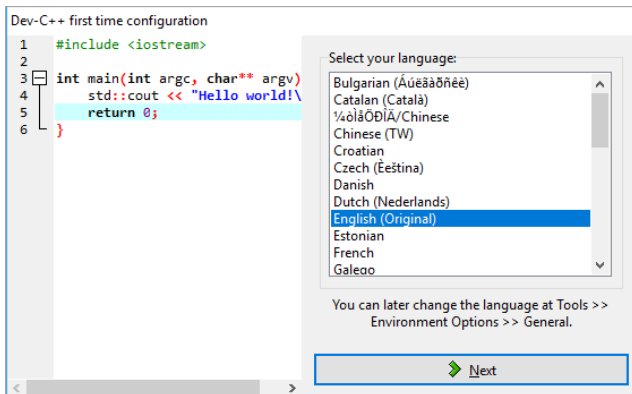


Figure 5: Language selection in Dev-C++.

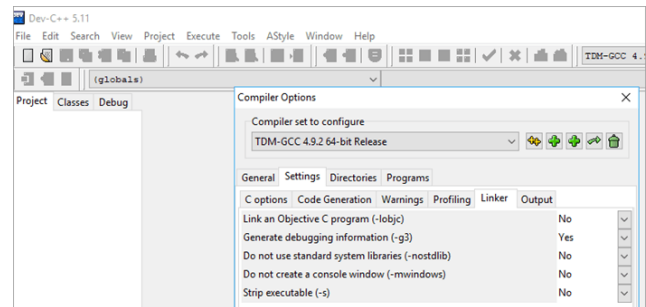


Figure 6: Compiler and linker configuration.

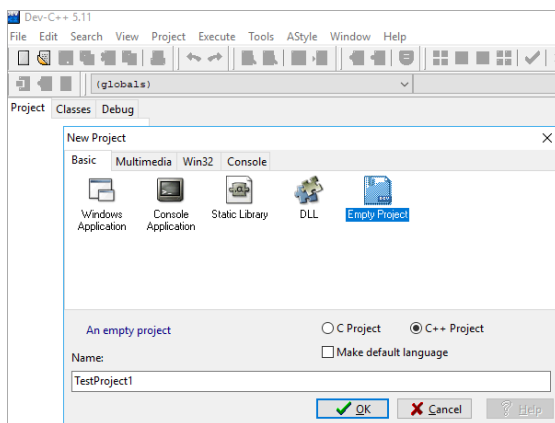


Figure 7: Creating a new C++ project.

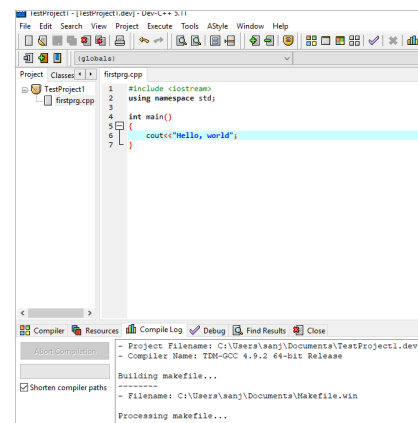


Figure 8: Editing a C++ source program.

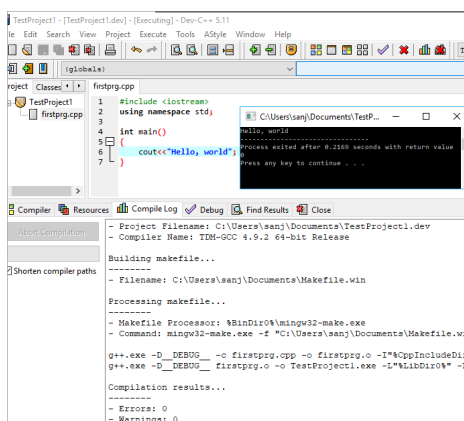


Figure 9: Compiling and running the program.

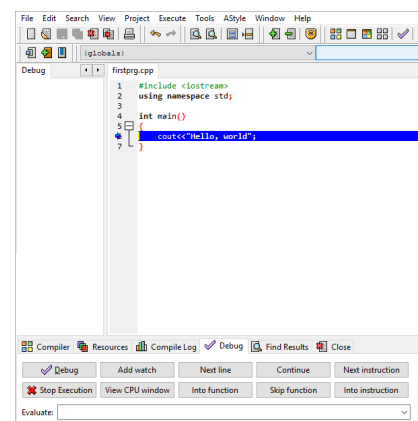


Figure 10: Debugging in the IDE.

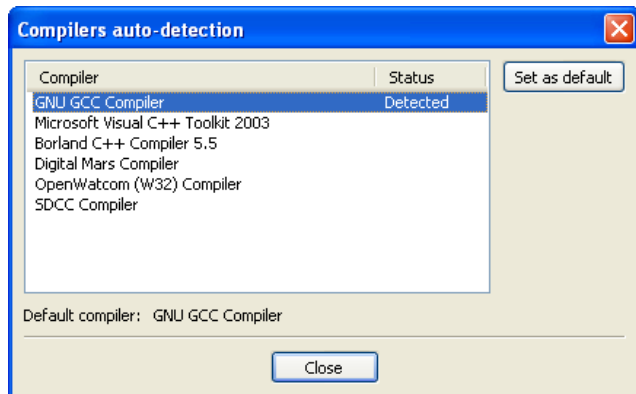


Figure 11: Compiler auto-detection in Code::Blocks.

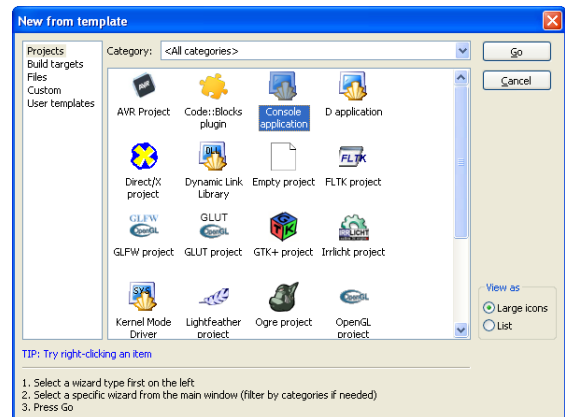


Figure 12: Creating a console application.

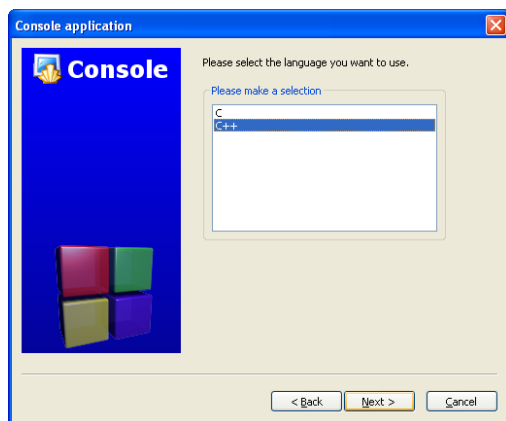


Figure 13: Selecting C or C++.

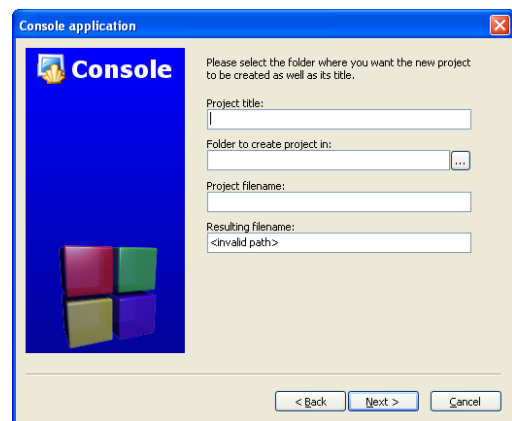


Figure 14: Specifying project location and name.

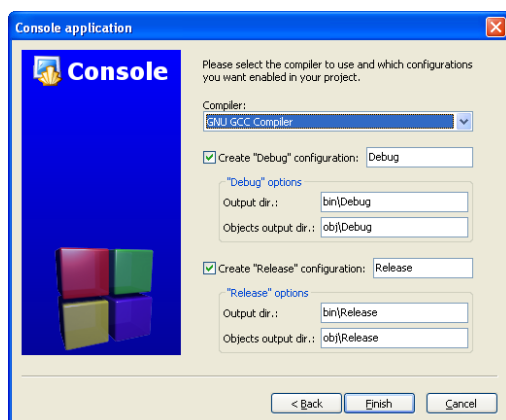


Figure 15: Compiler and build configuration.

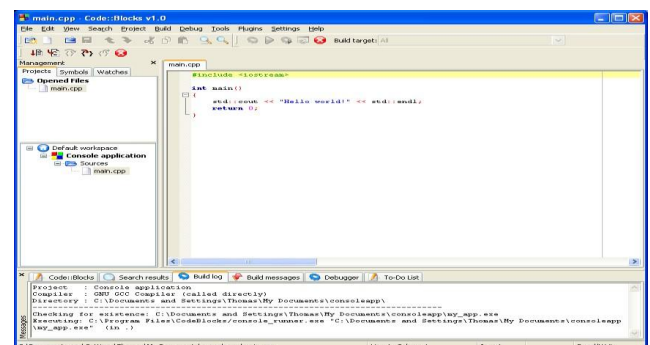


Figure 16: Code::Blocks project interface.

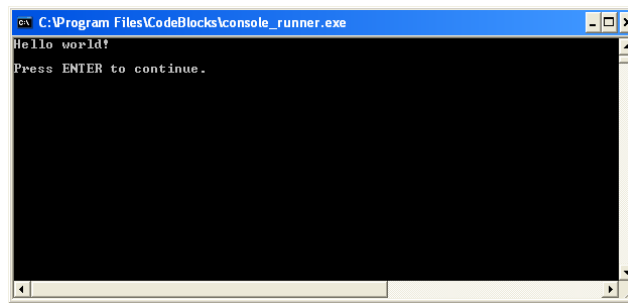


Figure 17: Console output from a compiled program.

WEEK #4**C++ Operators, Decision-Making and Control Statements****OBJECTIVES**

- ▶ To help the students in learning the concepts of C++.
- ▶ To help the students in learning about the different operators available in C++.
- ▶ To help the students in learning the different decision-making statements and control statements used in C++.

OUTCOMES

After completing this, the students would be able to

- ▶ Understand the concepts of C++.
- ▶ Understand the usage of various operators and their precedence in expression evaluation.
- ▶ Understand the usage of decision-making statements and control statements of C++ in real life applications.

PROBLEMS

Problem 1. Write a C++ program to check whether a number is even or odd using ternary operator.

Problem 2. Write a C++ program to perform the addition of two numbers without using + operator.

Problem 3. Write a C++ program to evaluate the arithmetic expression $((a + b / c * d - e) * (f - g))$. Read the values a, b, c, d, e, f, g from the standard input device.

Problem 4. A Fibonacci sequence is defined as follows: The first and second terms in the sequence are 0 and 1. Subsequent terms are found by adding the preceding two terms in the sequence. Write a C++ program to generate the first n terms of the sequence.

Problem 5. Write a C++ program to generate all the prime numbers between 1 and n, where n is a value supplied by the user.

Problem 6. A character is entered through keyboard. Write a C++ program to determine whether the character entered is a capital letter, a small case letter, a digit or a special symbol using if-else and switch case. The following table shows the range of ASCII values for various characters. Characters ASCII values A – Z: 65 – 90, a – z: 97 – 122, 0 – 9: 48 – 57 Special symbols 0 – 47, 58 – 64, 91 – 96, 123 – 127

Problem 7. Write a C++ program to find the roots of a quadratic equation.

Problem 8. Write a C++ program to check whether a given 3-digit number is an Armstrong number.

WEEK #5**One- and Two-Dimensional Arrays in C++****OBJECTIVES**

- ▶ To help the students in learning the concepts of arrays (1D and 2D) in C++.
- ▶ To help the students in implementing and handling 1D and 2D arrays with the help on various examples.
- ▶ To help the students in understanding various operations associated with the arrays.

OUTCOMES

After completing this, the students would be able to

- ▶ Handle arrays (1D and 2D) in the programs.
- ▶ Develop & implement simple real-life examples of arrays.

PROBLEMS

Problem 1. Write a C++ program to enter elements in the array and display the array elements.

Problem 2. Write a C++ program to find the sum of all the array elements.

Problem 3. Write a C++ program to find the length of the array.

Problem 4. Write a C++ program to find the second-largest integer in a list of integers.

Problem 5. Write a C++ Program to reverse the position of the array element (Hint: First element to the last element.)

Problem 6. Write a C++ program to perform the following: a. Addition of two matrices b. Multiplication of two matrices

Problem 7. Write a C++ program to count and display positive, negative, odd, and even numbers in an array.

Problem 8. Write a C++ program to merge two sorted arrays into another array in sorted order.

Problem 9. Write a C++ program to find the frequency of a particular number in a list of integers.

WEEK #6**Pointers and Pointer Operations in C++****OBJECTIVES**

- ▶ To help the students in learning the concepts of pointers in C++.
- ▶ To help the students implement and handle pointers with the help of various examples.
- ▶ To help the students understand various operations associated with pointers.

OUTCOMES

After completing this, the students would be able to

- ▶ Handle pointers in the programs.
- ▶ Develop & implement simple examples of pointers.
- ▶ Implement extensive use of pointers for memory, arrays, structures, and functions.

PROBLEMS

Problem 1. Write a C++ Program to Add Two Numbers Using a Pointer.

Problem 2. Write a C++ Example Program for swapping numbers Using Pointers.

Problem 3. Write a C++ Program to print the address of the Variable Using a Pointer.

Problem 4. Write a C++ Program for Increment and Decrement an Integer Using a Pointer.

Problem 5. Write a C++ Program to Print a String Using a Pointer.

Problem 6. Write a C++ program to concatenate two strings using pointers.

Problem 7. Write a program for reading elements using a pointer into an array and display the values using an array. i. Declare a set of elements. ii. Declare the pointer and initialize it to the first element's address of a set of elements(array). iii. Repeat the loop until the pointer reaches to the last element and displays each element.

Problem 8. Write a program using a pointer variable to the sum of n elements from the array.

Problem 9. Write a program for reading elements using a pointer into the array and display the values using an array.

Problem 10. Write a C++ program to reverse a string using pointers.

WEEK #7**Pointers and the extttthis Pointer****OBJECTIVES**

- ▶ To help the students implement and handle pointers with various examples.
- ▶ To help the students understand various operations associated with pointers.
- ▶ To help the students understand the concept of this pointer

OUTCOMES

After completing this, the students would be able to

- ▶ Handle pointers in the programs.
- ▶ Develop & implement simple examples of pointers.
- ▶ Implement extensive use of pointers for memory, arrays, structures, and functions.

PROBLEMS

Problem 1. Write a C++ Program to Count vowels in a String Using a Pointer

Problem 2. Write a C++ Program for the Length of a String Using a Pointer.

Problem 3. Write a C++ program using pointers to compute the sum, mean, and standard deviation of all elements stored in an array of n real numbers.

Problem 4. Write a C++ program to create three objects for a class named `pntr_obj` with data members such as `roll_no` & `name`. Create a member function `set_data()` for setting the data values and a `print()` member function to print which object has invoked it using the 'this' pointer.

Problem 5. Develop a C++ program to find the greatest of two numbers using this pointer, which returns the member variable.

Problem 6. Write a C++ program to implement a flight class with data members as flight number, source, destination, and fare. Write a member function to display the flight information using this pointer.

Problem 7. Write a C++ program to use this pointer and return the pointer reference.

WEEK #8**Strings and String Operations in C++****OBJECTIVES**

- ▶ To help the students in learning the concepts of strings in C++.
- ▶ To help the students implement and handle string operations with the help of various examples.

OUTCOMES

After completing this, the students would be able to

- ▶ Handle string manipulation in the programs.
- ▶ Develop & implement simple real-life examples of strings/string functions.

PROBLEMS

Problem 1. Write a C++ program that uses functions to perform the following operations:
i. To insert a substring into a given main string from a given position. ii. To delete n characters from a given position in a given string.

Problem 2. Write a C++ program to determine whether the given string is a palindrome.

Problem 3. Write a C++ program to find a string within a sentence and replace it with another string.

Problem 4. Write a C++ program that reads a line of text and counts all occurrences of a particular word.

Problem 5. Write a C++ program that displays the position or index in the string S where the string T begins, or 1 if S doesn't contain T.

WEEK #9**Functions, Recursion and Relational Database Queries****OBJECTIVES**

- ▶ To help the students implement and handle writing their own functions with the help of various examples.
- ▶ To help the students in learning Recursion.
- ▶ To help the students understand various operations associated with the function using an array.
- ▶ To learn how to write queries to create tables with constraints and insert and retrieve records.

OUTCOMES

After completing this, the students would be able to

- ▶ Understand how to access and use library functions in the programs.
- ▶ Understand Functions & Recursion.
- ▶ Understand the different constraints that can be applied to a table.
- ▶ Write DML and DDL-based queries.

PROBLEMS

Problem 1. Write C programs that use both recursive and non-recursive functions to find:

- a) The factorial of a given integer.
- b) To find the greatest common divisor of two given integers.

Problem 2. Write C programs that use recursive and non-recursive functions to solve towers of Hanoi problem.

Problem 3. Write a C++ program to print the transpose of a given matrix using a function.

Problem 4. Write a C++ program to swap two numbers by both call by value and call by reference mechanisms, using two functions swap_value() and swap_reference respectively, by getting the choice from the user and executing the user's choice using switch-case.

Problem 5. Write a C++ program to display all array elements using recursion.

Problem 6. Write a C++ program to find the sum of elements of an array using recursion.

Problem 7. Write a C++ program to find an array's maximum and minimum elements using recursion.

Problem 8. Consider the insurance database given below. The primary keys are made bold, and the data types are specified. PERSON (driver_id:string , name:string , address:string) CAR(regno:string , model:string , year:int) ACCIDENT(report_number:int , accd_date:date , location:string) OWNS(driver_id:string , regno:string) PARTICIPATED (driver_id: string, regno: string, report_number: int , damage_amount: int).

- ▶ Create the above tables by correctly specifying the primary keys and foreignkeys.
- ▶ Enter at least five tuples for each relation.
- ▶ Update the damage amount for the car with the specific regno in the accident with report number 12 to 25000.
- ▶ Add a new accident to the database.
- ▶ Find the total number of people who owned cars that were involved in accidents in 2008.
- ▶ Find the number of accidents in which cars belonging to a specific model were involved.

WEEK #10**Structures in C++ and Database Queries****OBJECTIVES**

- ▶ To help the students in learning the concepts of Structure in C++.
- ▶ To help the students implement structure with the help of various examples.
- ▶ To learn how to write queries to create tables with constraints and insert and retrieve records.

OUTCOMES

After completing this, the students would be able to

- ▶ Handle the problem based on the structure.
- ▶ Develop & implement simple real-life examples of structure.
- ▶ Understand the different constraints that can be applied to a table.

PROBLEMS

Problem 1. Write a C++ program that uses functions to perform the following operations:
i. Reading a complex number ii. Writing a complex number iii. Addition and subtraction of two complex numbers iv. Multiplication of two complex numbers. Note: Represent complex numbers using a structure.

Problem 2. Write a C++ program to compute the monthly pay of 100 employees using each employee's name, basic pay. The DA is calculated as 52% of the basic pay. Gross-salary (basic pay + DA). Print the employee's name and gross salary.

Problem 3. Create a Book structure containing book_id, title, author name and price. Write a C++ program to pass a structure as a function argument and print the book details.

Problem 4. Create a union containing 6 strings: name, home_address, hostel_address, city, state and zip. Write a C++ program to display your present address.

Problem 5. Write a C++ program to define a structure named D.O.B., which contains name, day, month and year. Using the concept of nested structures display your name and date of birth.

Problem 6. Consider the following database for a banking enterprise. BRANCH (branch_name:string , branch_city:string , assets:real) ACCOUNT(accno:int , branch_name:string , balance:real) DEPOSITOR(customer_name:string , accno:int) CUSTOMER (customer_name:string, customer_street:string , customer_city:string) LOAN (loan_number:int , branch_name:string , amount:real) BORROWER(customer_name:string , loan_number:int)

- a) Create the above tables by properly specifying the primary keys and foreign keys.
- b) Enter at least five tuples for each relation.
- c) Find all the customers who have at least two accounts at the main branch.
- d) Find all the customers who have an account at all branches in a specific city.
- e) Demonstrate how you delete all account tuples at every branch located in a specific city.

WEEK #11**Object-Oriented Programming, Classes and Objects****OBJECTIVES**

- ▶ To help the students in learning the concept of OOP.
- ▶ To help the students learn to write a sample program using OOP.
- ▶ To help the students understand the use of classes and objects.
- ▶ To help the students understand the concept of Access Specifier.

OUTCOMES

After completing this, the students would be able to

- ▶ Write, debug, and run a sample program using OOP.
- ▶ Create simple real-life applications.
- ▶ Understand the following points:
 - ▶ Class and object.
 - ▶ Creating a class and an object.
 - ▶ Defining a function in different ways.
 - ▶ Concept of Access Specifier like private, public, etc.

PROBLEMS

Problem 1. Write a program in C++ to display your name, Branch, and Year on the computer screen without using classes and objects. All information should be displayed in a separate line.

Problem 2. Write a menu-driven program in C++ to perform all basic arithmetic operations: addition, subtraction, multiplication, and division of two given values. The program receives two values and the required operation to be performed from the keyboard, and displays the particular result of the necessary operation.

Problem 3. Write a menu-driven program in C++ that receives a 4-digit integer value from the keyboard and performs following operations:

- i) Reverse of that number.
- ii) sum of number with its reverse.
- iii) sum of alternative digits (1 digit+3 digit and 2 digit+4 digit)

Problem 4. Write a menu-driven program in C++ to receive integer numbers and convert equivalent binary, octal, and hexadecimal numbers.

Problem 5. Write a menu-driven program in C++ to perform all basic arithmetic operations: addition, subtraction, multiplication, and division of two given values using functions and switch case. The program receives two numbers; the required operation to be performed from the keyboard, and displays the particular result of the required operation.

Problem 6. Define a class Bank Account to represent a bank account. Include the following members: Data Members: o Name of the depositor o Account Number o Type of account o Balance amount in the account Member Functions: o To assign an initial value o To deposit an amount o To withdraw an amount after checking

Problem 8. Write a C++ program to implement the Binary Search algorithm on a sorted array of integers and return the index of a target element.

Problem 7. Write a C++ program to implement the Bubble Sort algorithm to sort an array of integers in ascending order.

WEEK #12**Constructors, Destructors and Database Design****OBJECTIVES**

- ▶ To help the students understand the concept of a Constructor.
- ▶ To help the students understand the need for constructors and destructors in their programs.
- ▶ To help the students understand the use of copy and dynamic constructors.
- ▶ To learn how to write queries to create tables with constraints and insert and retrieve records.

OUTCOMES**After completing this, the students would be able to**

- ▶ Understand the use of the copy constructor and the dynamic constructor.
- ▶ Create a simple real-life application based on the constructors and destructors with static members.
- ▶ Understand the different constraints that can be applied to a table.

PROBLEMS

Problem 1. Write a program in C++ to demonstrate default constructor. Create a class with two data members in the private section. Define a default constructor to initialize these data members to their initial value and display these values with the help of a member function.

Problem 2. Write a program in C++ to demonstrate parameterized/constructor overloading. Create a class calculator that contains four data members. Initialize data members with different values using a parameterized constructor and perform various arithmetic operations over these values and display results on the computer screen.

Problem 3. Create a class called Triangle that stores the length of the base and height of a right triangle in two private instance variables. Include a constructor that sets these values. Define two functions. The first is hypo(), which returns the length of the hypotenuse. The

second is area (), which returns the area of the triangle.

Problem 4. Create a class for counting the number of objects created and destroyed within various blocks using constructor and destructors.

Problem 5. Create an inter University Database with the following relations. Include at least four attributes for each table.

- ▶ University
 - ▶ Department
 - ▶ Program
 - ▶ Course
 - ▶ Syllabus
 - ▶ Faculty(Teacher)
- a) Create the above tables and mention primary keys and foreign keys. Also, create a secondary index for each table.
- b) Insert at least five relevant records in each of the created tables.
- c) Write the following SQL queries based on the created database:
- i) List of Universities situated in Delhi.
 - ii) List of all Departments of AMU.
 - iii) Find the location of JNU.
 - iv) List of all Programs run by the University of Jammu.
- v) List of Universities that run the Program "MCA".
- vi) List of Courses of "MCA" run by AMU.
 - vii) List of Faculties specialized in "Information Security" across different universities.
 - viii) Syllabus of "Computer Architecture" of different Universities.

Problem 6. Write a C++ program to create a singly linked list and perform basic operations: insertion (at the beginning, end, and a given position) and deletion (at the beginning, end, and a given position).

WEEK #13**Operator Overloading, Type Conversion and Inheritance****OBJECTIVES**

- ▶ To help the students in learning operator overloading and the rules for overloading operators in C++.
- ▶ To help the students understand the overloading of the new and delete operators.
- ▶ To help the students learn the concept of Inheritance.
- ▶ To help the students understand the use of different forms of inheritance in their programs.

OUTCOMES

After completing this, the students would be able to

- ▶ Use binary operator overloading and understand the use of non-member functions in overloading.
- ▶ Create a simple real-life application based on Type conversion i.e., type conversion - basic type to class type, class type to basic type, class type to another class type.
- ▶ Know how to work/avoid ambiguity in multiple and multi-path inheritance.
- ▶ Create a simple real-life application based on a virtual base class and overriding member functions.

PROBLEMS

Problem 1. Declare a class Number that contains two data members value1 and value2 of the type of integer; define a constructor to give an initial value, and perform addition, subtraction, multiplication and division of these two numbers using operator overloading of +, -, *, / operator respectively [hint- binary operator overloading using member function]

Problem 2. Declare a class Number1 that contains two data members value1 and value2 of the type of integer; define a constructor to give an initial value, and perform addition, subtraction, multiplication and division of these two numbers using operator overloading of +, -, *, / operator respectively [hint- binary operator overloading using friend function]

Problem 3. Declare a class Number3 that contains a data member value of the type of integer, define a constructor to give an initial value, and perform unary minus, increment, and decrement this number using operator overloading of -, ++, -- operator respectively [hint- Unary operator overloading using member function]

Problem 4. Write a program to demonstrate explicit type conversion o from basic type to user-defined data type. o from User Defined data type to Basic data type data type.

Problem 5. Create a class publication which has title of book and writer's name. Create other class sales which account no. of sales for every month (up to 3 months)and then calculate total sales.

Problem 6. Write a program to demonstrate the following:

STAFF Teaching Non-Teaching Officer Regular Casual

Problem 7. Write a program to solve the ambiguity problem in inheritance where two different classes are inherited from a single base class and a new class is derivedfrom these two derived classes. How is this problem solved with the help of the virtual base class concept?

Problem 8. Write a C++ program to implement a stack using arrays and perform basic stack operations (push, pop, peek, and isEmpty).

Problem 9. Write a C++ program to implement a queue using arrays and perform basic queue operations (enqueue, dequeue, front, rear, and isEmpty).

WEEK #14**Binding, Virtual Functions, Streams and Exception Handling****OBJECTIVES**

- ▶ To help the students in learning the concept of binding in C++.
- ▶ To help the students learn how to write a sample program based on early binding and late binding in C++.
- ▶ To help the students understand the use of binding and virtual functions in their programs.
- ▶ To help the students understand the use of polymorphism.
- ▶ To help the students in learning the concept of streams in C++.
- ▶ To help the students understand the use of the hierarchy of file stream classes in their programs.
- ▶ To help the students understand the use of the Exception handling mechanism in their programs.
- ▶ To help the students understand the use of the throwing mechanism and catching mechanism

OUTCOMES**After completing this, the students would be able to**

- ▶ Write, debug, and run a sample program in C++.
- ▶ Differentiate between virtual functions and pure virtual functions.
- ▶ Create a simple real-life application based on polymorphism.
- ▶ Differentiate the way of reading/writing files, accessing records randomly, and updating files.
- ▶ Create a simple real-life application based on file handling.
- ▶ Understand hierarchical error handling: specific errors (think: derived classes) can be handled closer to where the error occurred; for instance, deep in the internals of some library. More general errors (think: base classes) can be handled higher up in the hierarchy; for instance, in client code that uses the library in C++.
- ▶ Differentiate try-catch blocks and throw.
- ▶ Create a simple real-life application based on the exceptional Handling statements.

PROBLEMS

Problem 1. Write a program to use 'this' pointer to find the elder from two person. Define a class Person to store age of the person. Define constructor/member function to give initial value to the data member age. And then define a function elder to compare ages of two different persons using this pointer to find out the elder person.

Problem 2. Create a simple "shape" hierarchy: a base class called Shape and derived classes called Circle, Square, and Triangle. In the base class, make a virtual function called draw (), and override this in the derived classes. Make an array of pointers to Shape objects that you create on the heap (and thus perform up casting of the pointers), and call draw () through the base-class pointers, to verify the behavior of the virtual function. If your debugger supports it, single-step through the code.

Problem 3. Write a small program to show the difference between calling a virtual function inside a normal member function and calling a virtual function inside a constructor. The program should prove that the two calls produce different results.

Problem 4. Write a program in C++ to calculate mean value of n numbers using friend function.

Problem 5. Write a program to accept five different numbers by creating a class called friend func1 and friend func2 taking 2 and 3 arguments respectively and calculate the average of these numbers by passing object of the class to friendfunction.

Problem 6. Write a program in C++ to display student's information using friend function.7# Write a C++ program to write text in the file. Read the text from the file from end of file. Display the contents of the file in reverse order.

Problem 8. Write a C++ program to count the no. of characters present in the file.

Problem 9. Create a class with a main () that throws an object of class Exception inside a try block. Give the constructor for Exception a String argument. Catch the exception inside a catch clause and print the String argument. Add a finally clause and print a message to prove you were there.

Problem 10. Write a C++ program to create a Binary Search Tree (BST) and perform basic operations: insertion, deletion, and search.